Normal Modes of the Market: Mean Reversion Across the Correlation Spectrum, Machine-Learned Mode Selection, and Regimes

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Abstract

We test whether the normal modes of the U.S. equity correlation matrix exhibit predictable mean reversion. In the overdamped limit of a coupled-oscillator market, the covariance matrix is the inverse of the stiffness matrix, so the lowest-variance eigenportfolios should relax fastest. Using 417 S&P 500 stocks, monthly walk-forward re-estimation and out-of-sample data from 2013 to 2024, we find that the modes the Marchenko–Pastur filter certifies as signal do not revert (Sharpe 0.07), while modes deep in the noise bulk do (Sharpe up to 0.87, $t = 2.74$), with significantly negative daily autocorrelation. Supervised models using eigenvalue separation, half-life, autocorrelation, eigenvector stability, volatility, portfolio concentration and regime features predict which modes will revert with out-of-sample AUC of 0.51–0.53: significant, and better than random selection, but not better than holding every mode. A two-state hidden Markov model reveals a strong regime asymmetry: reversion profits arise in high-volatility, high-correlation states and reverse in calm ones. A walk-forward strategy combining mode predictions with regime-based exposure reaches a net Sharpe ratio of 0.19 at 2 bp costs. That result depends on same-close execution and on an overlay whose form was chosen after observing the regime asymmetry. Overall, the normal-mode picture locates mean reversion correctly in the spectrum, but after costs, and once the specifications we tried are taken into account, it provides little tradable alpha.

1 Introduction

A system of coupled harmonic oscillators decomposes into normal modes: orthogonal coordinates that evolve independently, each with its own characteristic frequency. The eigen-decomposition of an equity correlation matrix has the same algebraic form. Its eigenvectors are portfolios with (in-sample) uncorrelated returns, and its eigenvalues are their variances. This paper asks whether the analogy can be made *predictive*. Do some normal modes of the market behave like displaced oscillators that are pulled back to equilibrium, and can we tell in advance which ones?

The question connects three literatures. Random matrix theory shows that most of the eigenvalues of a large sample correlation matrix are indistinguishable from noise, and the Marchenko–Pastur law gives a principled edge between signal and noise [4, 11, 16]. Statistical arbitrage trades the mean reversion of residual portfolios, typically residuals of stocks against principal-component or ETF factors [1, 5], and short-term reversal profits are known to vary strongly with market conditions

*University of Chicago. Code and data pipeline: https://github.com/besufikad-a/normal-modes-of-the-market.

[9, 12, 15]. Machine-learning methods have become a standard tool for extracting weak predictive signals from many correlated characteristics [7].

We make three contributions.

A physical prediction and a spectrum-wide test. We take the oscillator analogy to its overdamped, thermally driven limit, a multivariate Ornstein–Uhlenbeck process. In that model the return covariance and the stiffness matrix share eigenvectors and the *lowest*-variance modes are the fastest to revert. We then measure out-of-sample reversion for every mode of the spectrum. The modes above the Marchenko–Pastur edge, the conventional "signal", do not revert (Sharpe 0.07). Reversion is concentrated in the noise bulk (Sharpe 0.87 for modes 100–174, $t = 2.74$), and daily mode returns there have significantly negative autocorrelation. The direction of the physical prediction holds, but its quantitative form (half-life proportional to eigenvalue) does not.

Supervised prediction of mode quality. Using features computed only from each estimation window (eigenvalue separation, half-life, autocorrelation, eigenvector stability, volatility, portfolio concentration) plus market-regime variables, we train walk-forward logistic, gradient-boosting and random-forest classifiers to predict which candidate modes will revert profitably over the next month. The signal is statistically significant but small (out-of-sample AUC 0.51–0.53).

A walk-forward strategy with regime integration. We combine mode predictions with filtered hidden-Markov regime probabilities in a monthly re-estimated, market-neutral strategy, and evaluate returns, turnover, drawdowns and robustness to costs, execution delay and specification. Reversion is real before costs (gross Sharpe 0.81 when trading the full spectrum) but costly to harvest, with break-even costs of only 1.7–2.5 bp per unit turnover. ML selection beats random selection but not equal weighting. The largest improvement comes from regime timing: reversion pays in stressed markets and loses in calm ones.

Throughout, we report negative and null results alongside positive ones. The original specification in the accompanying code repository, a daily-return reversal on the top signal modes, has a gross Sharpe of -0.53 on our data, and a central purpose of this paper is to explain why.

2 Normal modes: from coupled oscillators to the return covariance

2.1 Coupled oscillators

Consider N unit masses with displacements $x \in \mathbb{R}^N$ coupled by springs with a symmetric positive-definite stiffness matrix K,

$$\ddot{x} = -Kx. \tag{1}$$

Writing $K = U\Omega^2 U^\top$ with U orthogonal and $\Omega^2 = \mathrm{diag}(\omega_1^2, \ldots, \omega_N^2)$, the normal coordinates $\xi = U^\top x$ decouple, $\ddot{\xi}_k = -\omega_k^2 \xi_k$, and every motion of the system is a superposition of N independent oscillations [6]. Two features of this textbook result carry over to markets: the decomposition is *orthogonal*, so normal coordinates can be traded as separate portfolios, and each mode has its own *time scale*, $2\pi/\omega_k$.

2.2 The overdamped analogue: an Ornstein–Uhlenbeck market

Prices are not inertial, so the relevant limit of (1) is the overdamped one with thermal forcing. Let x_t denote a vector of log-price deviations from some slowly moving equilibrium (for instance, idiosyncratic mispricings relative to a factor model) and suppose

$$\mathrm{d}x_t = -Kx_t\,\mathrm{d}t + s\,\mathrm{d}W_t, \tag{2}$$

a multivariate Ornstein–Uhlenbeck process [18] with isotropic noise of intensity s^2. The normal coordinates of K are again independent, now as scalar OU processes,

$$\mathrm{d}\xi_{k,t} = -\omega_k^2\,\xi_{k,t}\,\mathrm{d}t + s\,\mathrm{d}W_{k,t}, \tag{3}$$

each relaxing toward equilibrium at rate $\kappa_k = \omega_k^2$ (half-life $\ln 2/\omega_k^2$). The stationary covariance Σ solves the Lyapunov equation $K\Sigma + \Sigma K = s^2 I$, which gives

$$\Sigma = \frac{s^2}{2}\,K^{-1} \quad\Longrightarrow\quad \Sigma \text{ and } K \text{ share eigenvectors, with } \lambda_k = \frac{s^2}{2\,\omega_k^2}. \tag{4}$$

Equation (4) is equipartition: soft modes (small ω_k) fluctuate a lot, stiff modes fluctuate little. It also turns the covariance matrix into a map of relaxation speeds,

$$\kappa_k \;=\; \frac{s^2}{2\lambda_k}, \qquad \text{half-life}_k \;=\; \frac{2\ln 2}{s^2}\,\lambda_k. \tag{5}$$

In this idealized market, **the lowest-variance eigenportfolios are the fastest to mean-revert**, and the log half-life should increase one-for-one with $\log \lambda_k$. This is the prediction we take to the data. It is the opposite of the usual reading of a principal-component spectrum, in which the large eigenvalues are "signal" and the small ones are noise to be discarded.

Two caveats matter. First, (2) describes *levels*; we observe daily returns $r_t \approx \mathrm{d}x_t$, whose covariance is dominated by the noise term $s^2 I$ at short horizons. The eigenstructure of the return covariance reflects the drift term only through the (weak) autocorrelation of returns. The model is therefore a heuristic for where reversion should be found, not a structural model of the return covariance. Second, real return covariances are dominated by a market mode and a handful of sector modes whose eigenvalues are far larger than any OU drift could produce. We interpret these as common factors (non-stationary, $\omega_k \to 0$) and exclude the market mode from every strategy.

2.3 Mode portfolios

We work with the correlation matrix of returns standardized by their in-window volatilities σ_i. Its eigenvectors v_k (columns of V, ordered by descending eigenvalue $\lambda_0 \geq \lambda_1 \geq \dots$) define mode returns

$$r_{k,t} \;=\; \sum_i v_{ik}\,\frac{r_{i,t}}{\sigma_i}, \qquad \mathrm{Var}(r_k) \approx \lambda_k \text{ in-sample}, \tag{6}$$

which are the returns of the tradable portfolios with asset weights v_{ik}/σ_i [1]. The cumulative mode "price" $X_{k,t} = \sum_{s\leq t} r_{k,s}$ is the empirical counterpart of the normal coordinate ξ_k.

3 Separating signal from noise: Marchenko–Pastur

With T observations of N assets, sample eigenvalues are biased and noisy even when the true correlation matrix is the identity. In the limit $N, T \to \infty$ with $q = N/T$ fixed, the eigenvalues of a pure-noise correlation matrix follow the Marchenko–Pastur law [14],

$$\rho(\lambda) = \frac{\sqrt{(\lambda_+ - \lambda)(\lambda - \lambda_-)}}{2\pi q \lambda}, \qquad \lambda_\pm = (1 \pm \sqrt{q})^2, \tag{7}$$

with an additional point mass of weight $1 - 1/q$ at zero when $q > 1$. Eigenvalues above λ_+ are inconsistent with pure noise; following Laloux et al. [11] and Plerou et al. [16], we call them *signal*

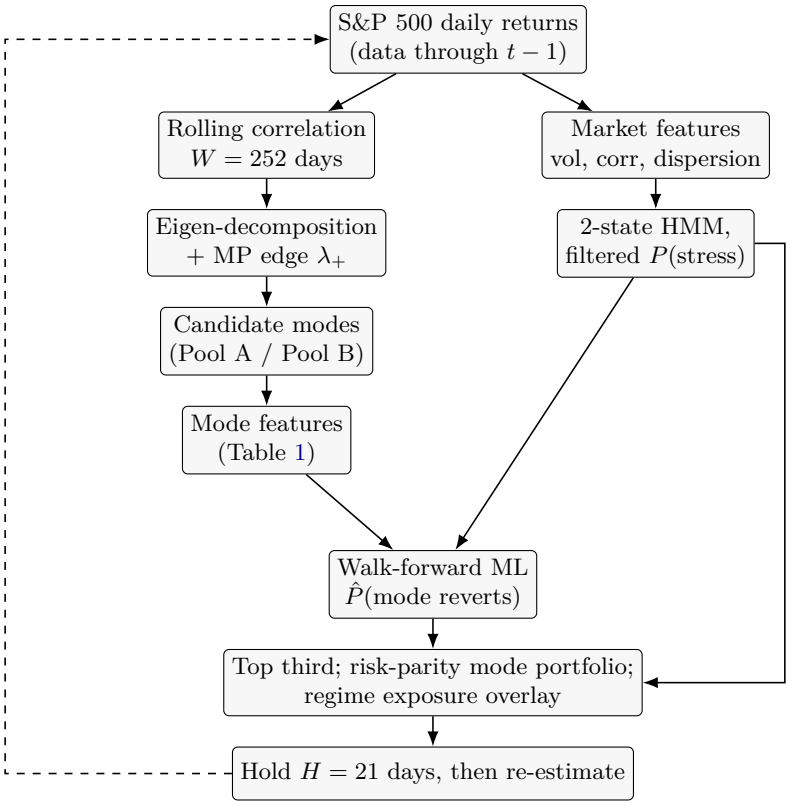


Figure 1: Pipeline. Each arrow uses only information available at the close before the episode starts. The dashed arrow marks the monthly walk-forward step.

modes and denote their number by n_s. In our rolling windows ($N = 417$, $T = 252$, $q \approx 1.65$) the edge is $\lambda_+ \approx 5.2$ and the sample has rank $T - 1 = 251$.

The MP edge is a statement about eigenvalues. For trading, the stability of eigen*vectors* matters more. Spiked-model theory [3] shows that a population eigenvalue just above the detection threshold produces a sample eigenvector that is only weakly aligned with its population counterpart, and below the threshold the alignment is asymptotically zero. Read literally, this argues for trading only modes well above λ_+. Read together with (5), it creates a tension that the paper is designed to resolve empirically. The modes that RMT certifies as real are the *soft* ones that should revert slowly. The modes that the OU picture says should revert fastest sit at the bottom of the spectrum, inside the noise bulk, where individual eigenvectors are unreliable. Even so, the subspace they span (the directions of least variance in the market) can be stable.

4 Data and walk-forward design

Data. We use daily dividend- and split-adjusted closing prices from Yahoo Finance for the 417 current S&P 500 constituents with a complete price history from January 2010 to December 2024 (3,772 trading days of log returns). Restricting to current constituents with full histories introduces survivorship bias (Section 9). Interior gaps of up to five days are forward-filled; no other cleaning is applied.

Episodes. Figure 1 summarizes the pipeline. The walk-forward is organized in *episodes*. At re-estimation day t, the correlation matrix is estimated on the trailing window $[t - W, t)$ with $W = 252$. Its eigenvectors and the asset volatilities σ_i are frozen and traded over the holding period $[t, t + H)$ with $H = 21$. The holding period equals the re-estimation step, so the episodes tile the sample without overlap: there are 168 episodes from January 2011. Everything recorded about the estimation window (features) uses data strictly before t, and everything recorded about the holding period (labels, PnL) uses data from t on. Because episode i's label is fully realized when episode $i + 1$ begins, a model trained on episodes $< j$ can score episode j with no overlap between training labels and test data.

Signal. On each trading day d of an episode, mode k's position is the negative, clipped z-score of its cumulative price relative to the previous $L = 60$ days,

$$s_{k,d} = -\operatorname{clip}\left(\frac{X_{k,d-1} - \bar{X}_k^{(L)}}{\operatorname{sd}(X_k)^{(L)}},\, -3,\, 3\right), \tag{8}$$

where the mean and standard deviation are taken over days $d - L, \ldots, d - 1$. This bets that a displaced normal coordinate is pulled back toward its recent equilibrium. The original specification in the code repository, a z-score of the daily mode *return* over 20 days, is a short-term reversal bet; we report it for comparison.

Label. The standalone forward PnL of mode k in an episode is $\pi_k = \sum_{d=t}^{t+H-1} s_{k,d}\, r_{k,d}$. The binary label is $y_k = \mathbf{1}\{\pi_k > 0\}$: did trading the mode's reversion signal make money over the next month? This ties "mean reversion" directly to the quantity a strategy can monetize. As a purely statistical check we also report the out-of-sample lag-1 autocorrelation of mode returns.

Candidate pools. Mode 0 is the market mode in every window (its overlap with the normalized equal-weight vector is 0.984 in the full sample) and is never a candidate. We fixed *Pool A*, modes 1–30, before seeing any out-of-sample result. It contains all signal modes (n_s ranges from 3 to 11 across episodes) plus the first part of the noise bulk, and so lets the models learn whether the MP edge matters. *Pool B*, every non-degenerate mode (1–249), was added after the spectrum-wide test in Section 7.2; results on Pool B should be read with that in mind.

5 Predicting which modes revert

5.1 Features

Every feature is computed from the estimation window only. Table 1 lists them by group. The groups follow the mechanisms discussed above: eigenvalue position (where the mode sits relative to the MP edge and its neighbours), estimated speed of reversion, serial dependence, eigenvector stability (whether the mode is the "same" portfolio from one month to the next), volatility regime of the mode, and portfolio concentration (whether the mode is a broad combination or a near-pair). We add the mode's trailing in-window reversion performance and, for the regime-aware models, the market-level state described in Section 6.

5.2 Models and protocol

We compare an ℓ_2-regularized logistic regression ($C = 0.1$, features standardized), histogram gradient boosting (depth 3, 150 iterations, learning rate 0.05, minimum leaf size 50), and a random forest (300

Table 1: Mode-level and market-level predictors. All are computed on the estimation window $[t-W, t)$ or, for regime variables, from data through the close of day $t-1$.

Group	Feature	Definition
Eigenvalue separation	`mode_index`	rank k
	`lam_over_mp`	λ_k/λ_+
	`log_lambda`	$\log \lambda_k$
	`eig_gap`	$\min(\lambda_{k-1}-\lambda_k,\ \lambda_k-\lambda_{k+1})/\lambda_k$
Half-life	`log_half_life`	log of $\ln 2/(-\ln b)$, b the AR(1) slope of X_k over the window (capped at 252 days)
Autocorrelation	`ac1`	lag-1 autocorrelation of r_k
	`vr5`	Lo–MacKinlay variance ratio VR(5) of r_k [13]
Stability	`stability`	$\max_j \lvert v_k^{(t)\top} v_j^{(t-H)}\rvert$ over the previous episode's modes
Volatility	`vol_ratio`	21-day volatility of r_k divided by $\sqrt{\lambda_k}$
Concentration	`participation`	$1/(N\sum_i v_{ik}^4)$, the effective fraction of assets in the mode
	`max_weight`	largest $\lvert v_{ik}/\sigma_i\rvert$ as a share of $\sum_i \lvert v_{ik}/\sigma_i\rvert$
Trailing	`trail_rev_sharpe`	Sharpe ratio of the mode's signal (8) over the last 63 window days
Regime	`p_stress`	filtered HMM stress probability
	`log_vol`	log 21-day realized volatility of the equal-weight market
	`avg_corr`	63-day average pairwise correlation
	`absorption`	λ_0/N, share of variance in the market mode [10]

trees, minimum leaf size 25). Hyperparameters were set once to conservative values and not tuned. We pre-specified the logistic model as primary because the signal-to-noise ratio of the problem is low. Models are trained on an expanding window of all (episode, mode) rows from episodes $< j$ and score episode j. The first prediction is made after 24 episodes (about two years), so the out-of-sample period starts on 7 January 2013. The logistic model is re-fit every episode and the tree ensembles every third episode. We report the out-of-sample AUC with 95% confidence intervals from a bootstrap that resamples whole episodes, which respects the cross-sectional dependence among modes in the same month.

6 Market regimes

Statistical-arbitrage returns are known to depend on the market environment. Reversion profits are typically higher when dispersion is high, and they can collapse in crowded deleveraging episodes [9]. We summarize the environment with three daily features computed from data through day d: the log 21-day realized volatility of the equal-weight market return; the 63-day average pairwise correlation; and the log 21-day average cross-sectional return dispersion. The average correlation is obtained without forming any matrix. For N unit-variance assets with average correlation $\bar{\rho}$, the cross-sectional mean of volatility-standardized returns has variance $\bar{\rho} + (1-\bar{\rho})/N$, which we invert.

A two-state Gaussian hidden Markov model [8] with diagonal covariances is fit by Baum–Welch [17]. To avoid look-ahead, the model is refit once a year on all data before 1 January, with features standardized using that training set's moments. The probabilities passed downstream are *filtered* (forward-algorithm) probabilities $P(\text{stress} \mid x_{1:d})$, never smoothed ones. States are ordered by mean volatility, so state 1 is always the high-volatility state, and the stress probability on the day before an episode starts enters both the ML models (as a feature) and the portfolio (as an exposure overlay).

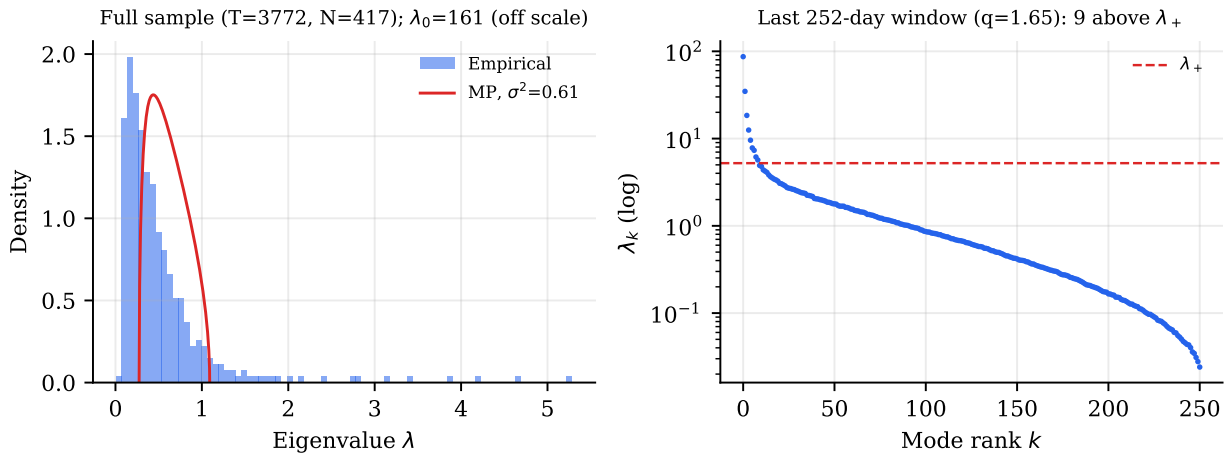


Figure 2: Left: full-sample eigenvalue density against the Marchenko–Pastur law, rescaled for the variance absorbed by the market mode. Right: eigenvalues of the last 252-day window on a log scale, with the edge λ_+ used in the walk-forward. Beyond rank $T - 1 = 251$ the eigenvalues are exactly zero.

7 Results

7.1 The spectrum

Figure 2 shows the familiar picture. In the full sample the market mode carries $\lambda_0 = 161$, or 39% of total variance, and its eigenvector is almost exactly the equal-weight portfolio (overlap 0.984). Once the bulk is rescaled for the variance absorbed by the market mode [11], the remaining eigenvalues sit close to the Marchenko–Pastur density, with 36 modes above the edge. In the rolling 252-day windows ($q = 1.65$) the edge is $\lambda_+ = 5.23$. Between 3 and 11 modes clear it, a handful of sector-like factors behind the market mode, and everything else is statistically indistinguishable from noise. (The walk-forward uses the unrescaled, conservative edge with $\sigma^2 = 1$. The rescaled edge admits about 13 modes per window, all inside Pool A, and changes none of the conclusions below.)

7.2 Where is the mean reversion?

Our first test uses no machine learning. In every episode we trade the standalone reversion signal (8) on *every* non-degenerate mode, normalize each mode's PnL by $\sqrt{\lambda_k}$, and average within buckets defined relative to that episode's own MP edge n_s. Table 2 and fig. 3 report out-of-sample (2013–2024), pre-cost results.

Three results stand out.

First, the modes that random matrix theory certifies as signal do not mean-revert in a tradable way. The level-reversion Sharpe ratio of the above-MP modes is 0.07 ($t = 0.26$). For the first 25 modes below the edge it is −0.00. The original repository's daily-return reversal signal does worse on the signal modes (−0.57, t =-1.96): these portfolios are sector-like factors that trend over days rather than revert.

Second, reversion lives in the bulk. The Sharpe ratio rises to 0.54 for modes $n_s + 25$ to 99, 0.87 for modes 100–174 ($t = 2.74$) and 0.70 for the lowest modes ($t = 2.39$). The same pattern, slightly stronger, holds for the return-reversal signal (1.10, $t = 4.07$). Purely statistically, every bucket has negative out-of-sample lag-1 autocorrelation. For the market and signal modes it is about −0.05 ($t \approx$-6.3), and in the bulk it is about −0.057 with t-statistics above 20 in absolute value. Mode

Table 2: Out-of-sample standalone reversion by position in the spectrum, 2013–2024, before costs. Buckets are defined relative to each episode's MP edge n_s. Sharpe ratios are annualized for an equal-risk portfolio of the modes in the bucket; t_{NW} uses Newey–West standard errors with 10 lags. The last two columns give the mean out-of-sample lag-1 autocorrelation of daily mode returns within an episode and its t-statistic across episodes.

	Level reversion		Return reversal		OOS lag-1 AC	
Modes	Sharpe	t_{NW}	Sharpe	t_{NW}	$\bar{\rho}_1$	t
Market ($k = 0$)	-0.09	-0.33	0.18	0.56	-0.066	-4.06
Signal, above MP ($1..n_s - 1$)	0.07	0.26	-0.57	-1.96	-0.047	-6.33
Edge bulk ($n_s..n_s + 24$)	-0.00	-0.02	-0.25	-0.71	-0.054	-12.05
Upper bulk ($n_s + 25..99$)	0.54	1.86	0.63	2.89	-0.057	-21.71
Middle bulk (100..174)	0.87	2.74	1.10	4.07	-0.057	-21.66
Lower bulk (175..249)	0.70	2.39	0.95	3.14	-0.056	-22.15

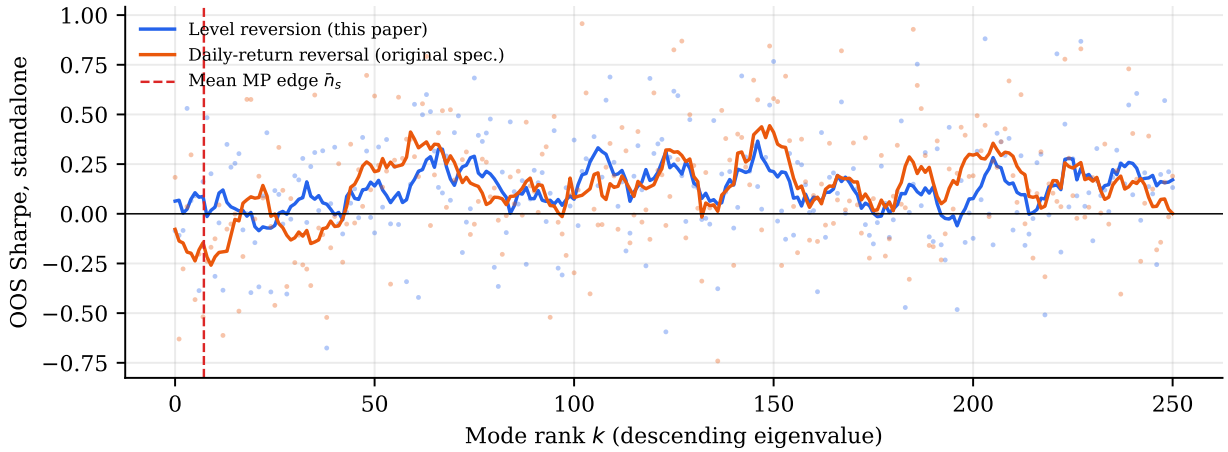


Figure 3: Out-of-sample Sharpe ratio of the standalone reversion signal for each mode rank (dots) and a 9-mode moving average (lines), 2013–2024, before costs. The dashed line marks the average MP edge.

returns at the bottom of the spectrum reverse day to day, as the overdamped picture of section 2 suggests. The reversal is economically small: a lag-1 autocorrelation of -0.06 explains well under 1% of next-day variance.

Third, the simple OU prediction for half-lives fails. Equation (5) predicts that the in-window half-life rises one-for-one with λ_k on a log scale. The fitted slope across all modes and episodes is -0.14, and the Spearman correlation is -0.30 (fig. 8 in the appendix). The 252-day AR(1) half-life of the level is a poor statistic for day-scale reversal, and the lowest-variance sample eigenvectors are partly artefacts of estimation noise, whose in-sample variance is biased downward [4]. The qualitative ordering of (4), with stiff directions reverting faster, shows up in out-of-sample *profitability*, not in in-sample half-lives.

These findings reframe the original thesis. The intuition that the "smallest signal modes" are the natural building blocks of a reversion portfolio is right about direction (small eigenvalues) but wrong about where to stop. The MP filter discards exactly the part of the spectrum where reversion is concentrated.

Table 3: Out-of-sample AUC for predicting $y = \mathbf{1}\{\pi_k > 0\}$, 2013–2024 (4,320 mode-months in Pool A, 35,856 in Pool B). Brackets: 95% intervals from a bootstrap over episodes. Heuristics rank modes by a single in-window statistic.

Pool	Model	Mode features	Mode + regime features
Modes 1–30	Logistic (L2)	0.523 [0.506, 0.540]	0.521 [0.501, 0.537]
	Gradient boosting	0.518 [0.502, 0.535]	0.518 [0.499, 0.538]
	Random forest	0.524 [0.508, 0.543]	0.528 [0.509, 0.545]
	Heuristic: $-\log$ half-life	0.512	
	Heuristic: $-\log \lambda$	0.504	
Full spectrum	Logistic (L2)	0.508 [0.500, 0.515]	0.514 [0.505, 0.522]
	Gradient boosting	0.506 [0.501, 0.512]	0.504 [0.497, 0.512]
	Random forest	0.507 [0.500, 0.513]	0.512 [0.505, 0.520]
	Heuristic: $-\log$ half-life	0.512	
	Heuristic: $-\log \lambda$	0.493	

Table 4: Feature-group importance for the primary (logistic, all-feature) model. "Solo AUC" trains on that group alone; "ΔAUC (drop)" is the change in out-of-sample AUC when the group is removed.

	Modes 1–30		Full spectrum	
Feature group	Solo AUC	ΔAUC (drop)	Solo AUC	ΔAUC (drop)
Eigenvalue separation	0.504	+0.003	0.507	-0.001
Half-life	0.510	+0.002	0.508	-0.001
Autocorrelation	0.522	-0.004	0.501	+0.001
Eigenvector stability	0.506	-0.000	0.498	+0.000
Volatility	0.512	-0.002	0.491	+0.000
Concentration	0.523	-0.005	0.490	-0.001
Trailing reversion	0.511	+0.001	0.497	+0.000
Regime	0.506	+0.002	0.512	-0.006

7.3 Can mode quality be predicted?

The base rate is high, with 62% of Pool A mode-months profitable out of sample. This reflects the negative skew of reversion payoffs: many small gains and occasional large losses. Table 3 reports out-of-sample AUCs.

Predictability is real but small. In Pool A every model clears 0.5. The logistic model with mode features reaches 0.523 [0.506, 0.540], and the random forest with all features reaches 0.528 [0.509, 0.545]. Gradient boosting is no better than the linear model, as expected when the signal-to-noise ratio is this low. In Pool B, where most rows are bulk modes that all revert to a similar degree, the AUCs fall to 0.504–0.514. The single-statistic heuristics score below the multivariate models: ranking by short half-life gives 0.512 in Pool A, and ranking by small eigenvalue gives 0.504, indistinguishable from chance.

The ablation in table 4 shows no single dominant group. The most informative groups on their own are portfolio concentration (broad modes revert more reliably than near-pairs) and autocorrelation; removing any one group changes the AUC by less than 0.006. Adding the regime features barely changes the AUC, but it does change the *level* of predicted probabilities. In Pool A the base rate is 65% in the stress state against 61% in the calm state, and the mean normalized forward PnL is negative in calm months. Timing, rather than cross-sectional selection, is where the regime information pays off (section 7.5).

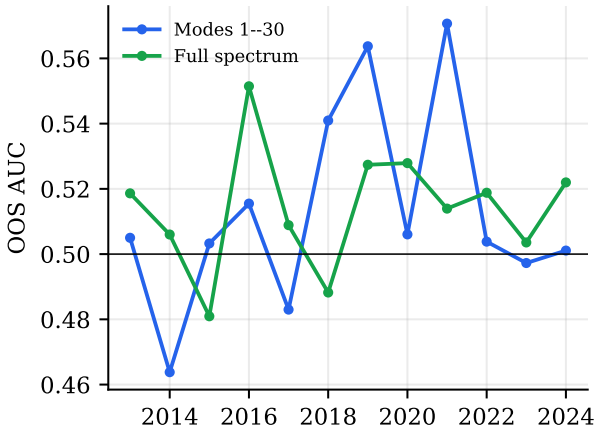


Figure 4: Out-of-sample AUC by calendar year for the primary model.

7.4 Regimes

The two HMM states are cleanly separated (fig. 5). In the most recent fit, the calm state has 10% annualized market volatility and an average pairwise correlation of 0.25, against 22% and 0.52 in the stress state. Both states are persistent (daily stay probabilities 0.991 and 0.981). The filtered stress probability exceeds one half on 29% of days: 78% of 2020, 92% of 2022 and 0% of 2017.

7.5 Walk-forward strategies

Construction. In each episode the selected modes are traded with signal (8), each scaled by $1/\sqrt{\lambda_k}$ for equal ex-ante risk. Mode positions are mapped to asset weights $x_i = \sum_k w_k\, s_k\, v_{ik}/(\sqrt{\lambda_k}\sigma_i)$ and rescaled daily to unit gross exposure. ML strategies select the top third of each episode's pool by predicted probability. Costs are 2 bp per unit of one-way turnover. All strategies are evaluated on the same period (7 January 2013 – 30 December 2024).

Regime overlay. The episode-level exposure is $e_j = (1 - p_j)g_{\text{calm}} + p_j g_{\text{stress}}$, where p_j is the filtered stress probability before the episode and $g_s = \text{clip}(\widehat{SR}_s / \max_{s'} \widehat{SR}_{s'}, 0, 1)$. Here $\widehat{SR}_s$ is the strategy's own past gross Sharpe ratio on days classified to regime s, using only completed episodes. The best regime so far is traded at full size and a regime with a negative past Sharpe is not traded. *Disclosure:* our pre-specified overlay could only de-risk in the stress state. It was never active, because the strategy's past stress-state Sharpe always exceeded its calm-state Sharpe. The symmetric rule above replaced it after we saw this. The rule is still strictly walk-forward, but its form was informed by the data, and its results should be discounted accordingly.

The original specification fails. The original repository strategy (daily-return reversal on the top 20 signal modes, raw-return projection) has a gross Sharpe of -0.53. With daily turnover of 1.36 times its book it loses heavily after costs (net Sharpe -1.53). Its failure is predicted by table 2: it trades the part of the spectrum that does not revert, with a signal that is negative there.

Gross reversion is real, but small relative to turnover. Trading every Pool B mode with equal risk earns a gross Sharpe of 0.81, and does so with almost no market exposure. Per unit of gross exposure, however, the return is only 1.8% a year against daily turnover of 0.42, a break-even

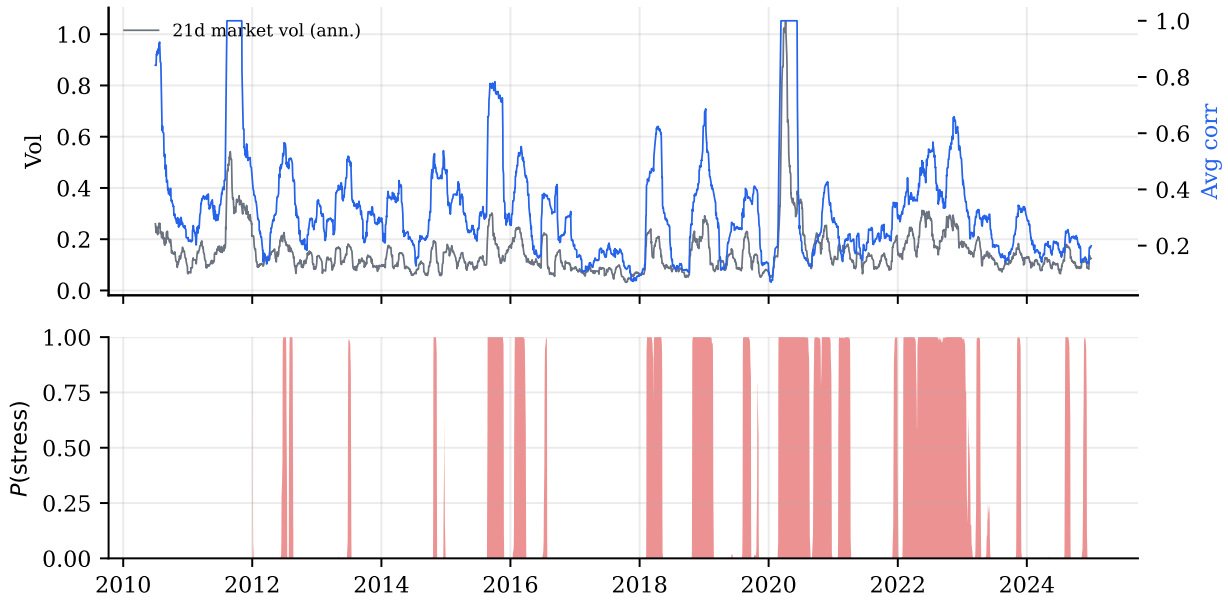


Figure 5: Regime features (top) and the walk-forward filtered stress probability (bottom).

cost of 1.70 bp. At 2 bp the net Sharpe is −0.14. Pool A is weaker throughout (gross 0.16). The long-short book is extremely diversified, so its volatility per unit gross is low (2.2%), and a levered implementation would scale return and volatility together. Costs, not risk, are the binding constraint.

ML selection beats random selection, not diversification. In Pool B, the ML-selected top third has a gross Sharpe of 0.85. A randomly selected top third of the same size averages −0.71 net (95th percentile −0.46), and the ML strategy beats every one of 200 random draws ($p < 0.005$). But the ML strategy does not beat simply trading all candidates. The Sharpe difference against equal weight is +0.03 (bootstrap $p = 0.46$), and in Pool A it is negative (−0.30). Holding all modes diversifies away more noise than a weak classifier can add by choosing among them. The half-life heuristic is significantly *worse* than equal weight in Pool B (−0.52, p for improvement = 1.00).

Regimes matter most. Reversion makes money in the stress state and loses in the calm state, for every strategy and both pools. For the Pool B equal-weight portfolio the Sharpe ratio is 0.79 in stress and −1.05 in calm. This matches the evidence that short-term reversal profits are a compensation for liquidity provision that rises with volatility [9, 15]. The overlay, which learns this asymmetry walk-forward, raises the net Sharpe of the equal-weight Pool B portfolio to 0.16 (difference +0.31, $p = 0.01$). Combined with ML selection it reaches 0.19 (+0.33, $p = 0.09$) and reduces the maximum drawdown from −10.7% to −6.7%. Part of the gain comes from lower turnover, since the overlay often trades at reduced size. The gross Sharpe also rises, to 0.95.

7.6 Robustness

Figure 7 and table 6 summarize the sensitivity analysis. Net results are fragile to costs: at 5 bp every strategy has a clearly negative Sharpe (Pool B equal weight: −1.57), and at zero cost the Pool B equal-weight Sharpe is 0.81. Adding one day of execution delay roughly halves the gross

Table 5: Walk-forward strategy performance, 7 January 2013 – 30 December 2024. Net of 2 bp per unit turnover unless labelled gross. Turnover is average daily one-way turnover as a fraction of gross exposure. Break-even is the cost in bp per unit turnover at which the net return is zero.

Strategy	Pool	Ann. ret.	Ann. vol	Sharpe	Gross SR	Max DD	Turnover	Break-even
Original repo specification	–	-10.5%	6.9%	-1.53	-0.53	-125.8%	1.36	-1.06
MP signal modes, equal weight	1–30	-0.7%	5.2%	-0.14	0.11	-20.2%	0.26	0.88
All candidates, equal weight	1–30	-0.8%	3.8%	-0.20	0.16	-22.7%	0.27	0.88
Heuristic: shortest half-life	1–30	-1.1%	2.5%	-0.46	0.14	-15.1%	0.30	0.47
ML: mode features	1–30	-0.8%	2.7%	-0.29	0.24	-15.7%	0.29	0.91
ML: mode + regime features	1–30	-1.3%	2.5%	-0.50	0.07	-15.2%	0.29	0.25
ML + regime + exposure overlay	1–30	-0.4%	2.0%	-0.21	0.18	-9.2%	0.15	0.93
Equal weight + exposure overlay	1–30	-0.3%	3.2%	-0.09	0.14	-14.4%	0.14	1.26
All candidates, equal weight	Full	-0.3%	2.2%	-0.14	0.81	-10.7%	0.42	1.70
Heuristic: shortest half-life	Full	-1.1%	1.6%	-0.67	0.67	-13.0%	0.43	1.00
ML: mode features	Full	-0.5%	1.9%	-0.25	0.74	-11.6%	0.37	1.50
ML: mode + regime features	Full	-0.2%	1.9%	-0.12	0.85	-10.8%	0.36	1.76
ML + regime + exposure overlay	Full	0.3%	1.6%	0.19	0.95	-6.7%	0.24	2.49
Equal weight + exposure overlay	Full	0.3%	1.9%	0.16	0.89	-6.2%	0.28	2.44

Sharpe ratio in Pool B (from 0.81 to 0.42 for equal weight, and from 0.85 to 0.33 for ML; table 6). A large share of the measured reversion is therefore a one-day effect, consistent with close-to-close bid–ask bounce and short-lived liquidity shocks. Strategies that cannot trade at the close would capture much less of it. In Pool B the gross Sharpe ratio stays positive in every re-specification (window, lookback, holding period, selection fraction, model), but the ML strategy beats equal weight net of costs only in a few of them (e.g. $W = 504$ or selecting the top half). In Pool A gross Sharpe ratios are near zero throughout. Gross performance is strongest in 2013–2016 and 2020–2021 and weak in 2022–2024. Gradient boosting and random forests produce worse strategies than the logistic model, despite similar AUCs.

8 Discussion

What the normal-mode analogy gets right. Taken seriously as an overdamped, thermally driven system, the coupled-oscillator picture predicts that the *lowest*-variance eigenportfolios relax fastest. Out of sample, that ordering holds. Reversion profits and negative autocorrelation are concentrated at the bottom of the spectrum, and the market and sector modes at the top trend rather than revert. The analogy is therefore a useful guide to where to look, and it inverts the usual PCA habit of treating the leading components as the interesting ones.

What it gets wrong. The quantitative prediction, a half-life proportional to λ_k, is not supported. The reversion we measure is mostly a one-day effect that does not survive an extra day of execution delay. That is more naturally read as liquidity provision and microstructure noise [12, 15] spread over many small-variance combinations of stocks than as slow relaxation toward an equilibrium.

The Marchenko–Pastur filter is the wrong filter for this purpose. RMT tells us which eigenvalues are distinguishable from noise, and that is the right question for covariance cleaning [4, 11]. For reversion trading, the relevant object is the low-variance *subspace*. Individual bulk eigenvectors are unstable, but projecting onto many of them and trading them together diversifies away their individual instability. This explains why equal-weighting the whole bulk beats every selection rule we tried.

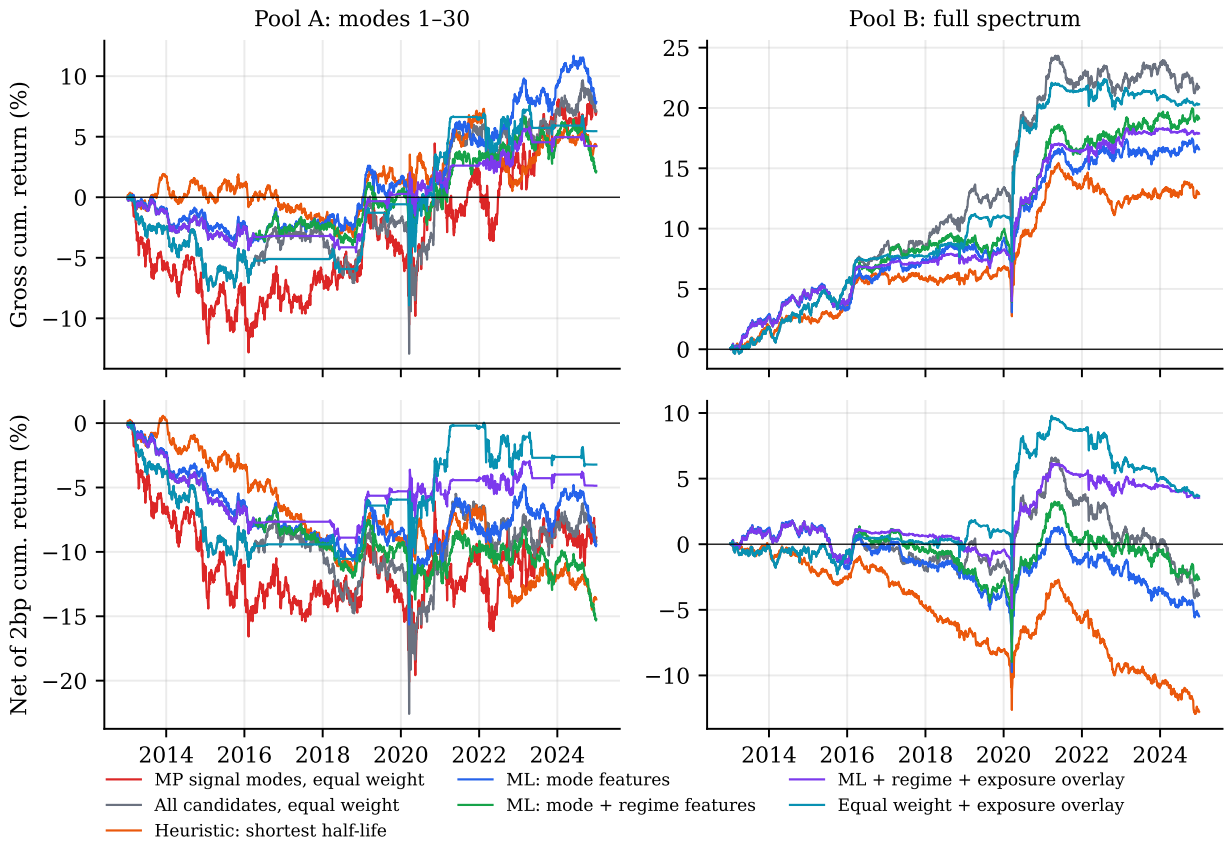


Figure 6: Cumulative log returns per unit gross exposure, before (top) and after (bottom) 2 bp costs.

Machine learning adds little; timing adds more. With AUCs of 0.51–0.53, the classifiers do find structure. They are significantly better than random selection, and stability, concentration and autocorrelation carry information. But their edge is smaller than the diversification cost of trading only a third of the pool. Regime information is more valuable as a *timing* signal for the whole book than as a feature for ranking modes. The regime asymmetry is large: reversion is unprofitable in calm markets and profitable in stressed ones. Our overlay exploits it walk-forward, but its form was chosen after seeing this asymmetry.

Multiple testing. We report every strategy and variant we ran. Across them, the best net Sharpe ratio is below 0.2 over twelve years, so none would survive a deflated-Sharpe adjustment for the number of trials [2]. The robust findings are the cross-spectrum pattern (table 2), which involves no selection, and the regime asymmetry, which holds in every specification.

9 Limitations

- **Survivorship.** The universe is the current S&P 500 restricted to stocks with complete 2010–2024 histories (417 names). This removes delisted and acquired firms. Survivorship typically flatters long-only returns more than market-neutral reversion, but it can also understate the tail risk of a reversion book, which loses when a stock falls and never comes back. Point-in-time constituents (e.g. CRSP) would fix this.

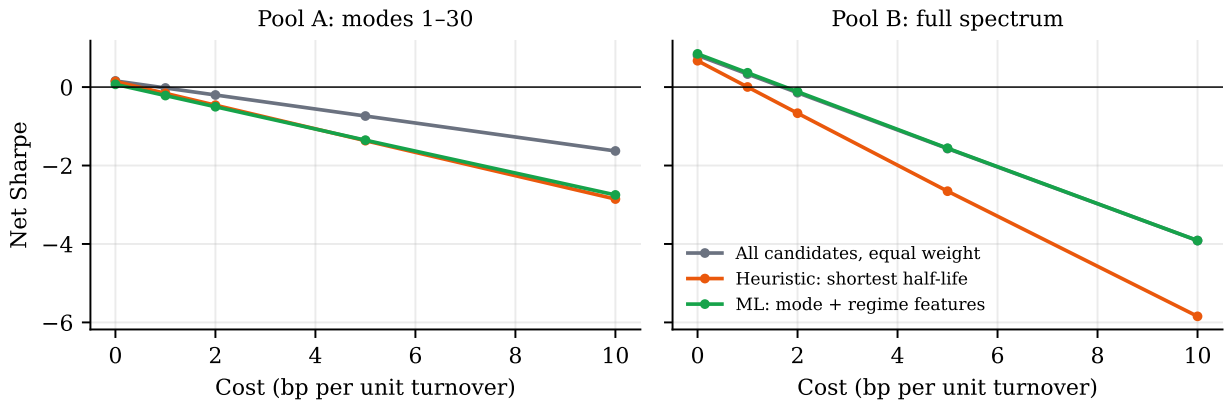


Figure 7: Net Sharpe ratio as a function of the cost per unit turnover.

- **Execution.** Signals use closing prices and are traded at the same close. The execution-delay test shows that this assumption matters a great deal. The cost model is a flat charge per unit turnover, with no market impact, borrow costs or financing.
- **Estimation.** We use sample correlations with a fixed 252-day window. Shrinkage or rotationally invariant estimators [4], or eigenvector tracking across windows rather than re-estimating from scratch, could make individual modes more stable and selection more valuable.
- **Label.** The binary profitability label discards magnitude, and the payoff is negatively skewed. Regression or quantile targets, or a direct portfolio objective, might use the weak signal more efficiently.

10 Conclusion

We set out to test whether the low-eigenvalue normal modes of the equity correlation matrix mean-revert predictably, and whether supervised learning and regime detection can turn that into a statistical-arbitrage strategy. On 417 S&P 500 stocks over 2013–2024, out of sample, the answers are as follows. Yes, mode portfolios mean-revert, but at the bottom of the spectrum rather than in the Marchenko–Pastur signal modes, and mostly at a one-day horizon. Only weakly can it be predicted which modes will revert: AUC 0.51–0.53, better than chance and better than random selection, but not better than holding everything. Market regimes predict *when* reversion pays far better than mode features predict *which* modes pay. After realistic costs, the best walk-forward implementation earns a small positive Sharpe ratio, 0.19, that depends on a data-informed overlay and on same-close execution. The coupled-oscillator analogy is a good map of where reversion lives in the spectrum. It is not, on this evidence, a source of tradable alpha for a daily close-to-close strategy.

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Table 6: Robustness of the primary ML strategy (logistic, mode + regime features, top third, no overlay) and the equal-weight benchmark. Net Sharpe at 2 bp, with gross Sharpe in parentheses. Rows below the rule re-build the episodes and re-train the models; sub-period rows split the baseline run.

	Pool A: modes 1–30			Pool B: full spectrum		
Test	ML	EW	AUC	ML	EW	AUC
Baseline (W=252, L=60, H=21)	-0.50 (0.07)	-0.20 (0.16)	0.521	-0.12 (0.85)	-0.14 (0.81)	0.514
Sub-period 2013–2016	-1.27 (-0.42)	-0.92 (-0.37)	–	0.04 (1.48)	0.04 (1.61)	–
Sub-period 2017–2019	-0.20 (0.51)	-0.29 (0.23)	–	-0.74 (0.66)	-0.34 (1.05)	–
Sub-period 2020–2021	-0.05 (0.34)	0.30 (0.52)	–	0.44 (1.03)	0.56 (1.09)	–
Sub-period 2022–2024	-0.65 (-0.11)	-0.19 (0.19)	–	-0.49 (0.47)	-1.20 (-0.12)	–
Select top 20%	-0.63 (-0.04)	-0.20 (0.16)	–	-0.25 (0.82)	-0.14 (0.81)	–
Select top 50%	-0.42 (0.07)	-0.20 (0.16)	–	-0.08 (0.85)	-0.14 (0.81)	–
Model: GBM	-0.65 (-0.16)	-0.20 (0.16)	–	-0.56 (0.41)	-0.14 (0.81)	–
Model: RF	-0.36 (0.15)	-0.20 (0.16)	–	-0.54 (0.64)	-0.14 (0.81)	–
Execution delay +1 day	-0.53 (0.01)	-0.27 (0.10)	0.514	-0.74 (0.33)	-0.59 (0.42)	0.508
Window $W = 126$	-0.60 (-0.01)	-0.24 (0.15)	0.521	-0.63 (0.35)	-0.28 (0.36)	0.516
Window $W = 504$	-0.58 (-0.10)	-0.10 (0.24)	0.510	-0.13 (0.87)	-0.31 (0.83)	0.519
Lookback $L = 20$	-0.70 (0.19)	-0.43 (0.24)	0.507	-0.97 (0.64)	-0.36 (0.97)	0.513
Lookback $L = 120$	-0.66 (-0.13)	-0.21 (0.03)	0.519	-0.82 (0.55)	-0.13 (0.60)	0.521
Holding $H = 10$	-0.31 (0.45)	-0.07 (0.30)	0.524	-1.22 (0.64)	-0.44 (0.72)	0.518
Holding $H = 42$	-0.35 (0.17)	-0.26 (0.09)	0.510	-0.24 (0.63)	-0.12 (0.66)	0.517

selection bias, backtest overfitting, and non-normality. *Journal of Portfolio Management*, 40 (5):94–107, 2014.

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A Additional figures

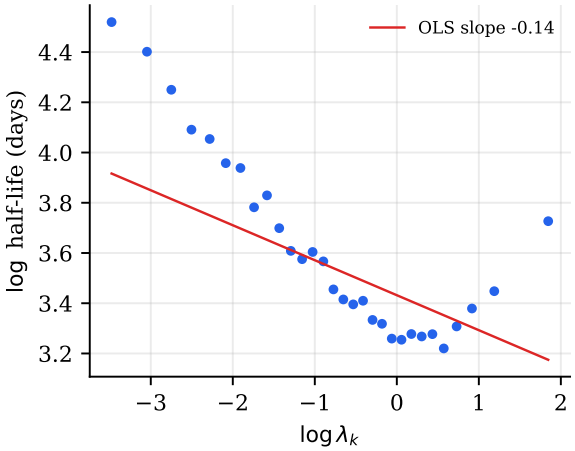


Figure 8: In-window AR(1) half-life of the cumulative mode price against $\log \lambda_k$, all modes and episodes (medians in 30 eigenvalue bins). The OU model (5) predicts slope $+1$; the fitted slope is -0.14. 10% of mode-windows show no detectable reversion ($b \geq 1$) and are excluded from the fit.

B Implementation details

HMM estimation. Baum–Welch uses the scaled forward–backward recursions of Rabiner [17]. It is initialized by splitting the sample into volatility terciles (halves, for two states), with a transition matrix of 0.98 on the diagonal, and runs to a log-likelihood tolerance of 10^{-6}. Variances are floored at 10^{-4} in standardized units. The first fit uses data from 2010 to 2011 and produces probabilities from January 2012.

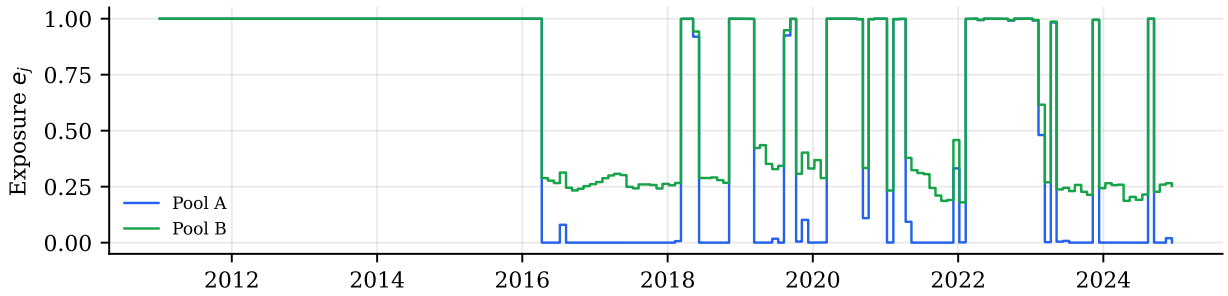


Figure 9: Walk-forward regime-overlay exposure e_j for the ML strategies. Average exposure is 0.52 (Pool A) and 0.65 (Pool B).

Leakage checks. The test suite (`tests/`) perturbs future data and verifies that nothing computed for the past changes. It covers the signal, every episode feature, the HMM filter, the walk-forward HMM refits, and the walk-forward classifier, which is checked to be invariant to future labels. It also checks that a single-mode portfolio reproduces the sign of that mode's standalone PnL, and that the HMM recovers known states on synthetic data.

Reproduction. `python scripts/run_study.py` downloads the data (cached), runs every stage, and writes all figures and tables in this paper together with `output/results.json`. `python scripts/make_macros.py` then regenerates every number quoted in the text. A full run takes about ten minutes on a laptop.